

The Impartial Realizer Theorem: A Pass-Free Quadratic Arena

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Abstract

The weighted-source WITT-FIFO theorem of the companion flagship paper realizes $\{Q = 0\}$ as a normal-play P -set for every quadratic refinement Q of every alternating form on $\mathbb{F}_2^n$, under the optimal $(w_0, c) = (1, 1)$ access contract. We prove that the arena can be made impartial. Delete PASS and allow a ko-protected FIFO front to close once the untouched set is empty. Ko still forces every matching edge to overlap, while the core now has exactly two moves per loaded coin. One seat-symmetric tail move at charge one therefore compiles the forced charge to ordinary normal play: the first player forces the tail when $Q(x) = 1$, and the second player forces its absence when $Q(x) = 0$. The construction is uniform over all finite dimensions, alternating forms, refinements, and inputs, and uses at most one singleton query per transition.

1 Statement

Throughout, $V = \mathbb{F}_2^n$, B is an alternating bilinear form on V , and Q is a quadratic refinement of B , i.e. $Q(u+v) = Q(u) + Q(v) + B(u, v)$ with $Q(0) = 0$. The access contract F1–F3/N1–N2, the loading map, and the realization theorem are those of the flagship paper [1]: statics are refinement-blind (F1), each move predicate and transition observes Q through at most c oracle queries at points of Hamming weight at most w_0 , with (w_0, c) independent of dimension (F2), the semantics is one declared outcome convention (F3), and exactness holds uniformly over the full torsor of refinements of every form in the family (N1) within the stated budget (N2).

Theorem 1 (impartial uniform realizer). **[PROVED]** *There is one impartial normal-play rule satisfying F1–F3/N1–N2 at $(w_0, c) = (1, 1)$ whose loaded root at x is P exactly when $Q(x) = 0$, uniformly over every quadratic refinement Q of every alternating form B in every finite dimension.*

2 The proved fence

Three flagship results show what the theorem must accomplish.

Theorem 2 (coin-turning linear ceiling). **[PROVED]** *[1, Thm. A] Every Winning Ways coin-turning game has configuration P -set equal to the kernel of an $\mathbb{F}_2$ -linear nimber-valued map; no coin-turning P -set is a quadric of positive polar rank.*

An impartial realizer therefore cannot be a disjunctive sum of coin-turning heaps, and more generally cannot present its positions as configurations whose Grundy value is XOR-additive over independent components. Whatever forces the quadric must entangle the coordinates dynamically, as the FIFO queue does.

Theorem 3 (abelian obstruction). **[PROVED]** [1, Lem. abelian, Thm. B] *In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular no commutative game-value monoid – Sprague–Grundy values under disjunctive sum, misère quotients, turn-parity counters – carries a quadratic refinement with $B \neq 0$.*

This does *not* exclude the theorem. It excludes factorizations: an impartial candidate cannot read Q off any commutative value quotient of its positions. The refinement must enter intensionally, through the move relation. Impartial play retains the one asymmetry the flagship identifies as structurally available [1, Rem. screen] – the first-/second-player asymmetry of whose turn it is – even though its move *sets* are equal. Normal play is precisely the reading of that asymmetry.

Theorem 4 (transcript-span bound). **[PROVED]** [1, transcript-span theorem] *Any transcript-stable exact rule observes refinement values at directions spanning the input; in the original frame, $\text{weight} \leq w$ observations number at least $\text{wt}(x)/w$.*

The bound is seat-blind: an impartial realizer at $w_0 = 1$ still needs the full singleton support $\{e_j : x_j = 1\}$. No information discount is available for symmetry.

3 The pass-free rule

Use the flagship’s deterministic B -only adapted basis f_i , active coordinates y_i , public corrections $p_i = P_B(f_i)$, and source pair S_j^0, S_j^1 for every $x_j = 1$. Let G_0 be the public potential matching consisting of every loaded Witt edge and every loaded source pair, regardless of the source weight $q_j = Q(e_j)$.

The core state is $(U, L, \text{ko}, \sigma)$, with untouched coins U , FIFO queue L , and accumulated charge σ . OPEN is unchanged. Delete PASS, and declare CLOSE legal exactly when $L \neq \emptyset$ and

$$\text{ko} = 0 \quad \text{or} \quad U = \emptyset.$$

At a drained state there is one move to a fixed optionless sink exactly when $\sigma = 1$. Every move is seat-symmetric.

4 Proof

Fix either designated seat. Its public policy is deterministic: while $U \neq \emptyset$, never CLOSE; from an empty queue OPEN the least untouched coin; otherwise OPEN the untouched G_0 -mate of the front if it exists, and OPEN the least untouched coin if it does not. Once $U = \emptyset$, CLOSE.

Call a checkpoint handed to the opponent safe when every legal CLOSE has zero live G_0 -degree. An OPEN onto an empty queue sets ko. If untouched coins remain, CLOSE is illegal; if none remain, every live degree is zero. From a nonempty queue, opening the front’s public mate removes its unique possible live neighbour, while opening any coin when no mate remains preserves degree zero. Thus every policy move hands over a safe checkpoint. An opponent OPEN returns the turn; an opponent CLOSE has zero live degree and returns the turn. The newly legal ko-CLOSE is harmless because its legality forces $U = \emptyset$. Induction on $2|U| + |L|$ proves that every CLOSE along a policy-consistent play has zero public live degree. The empty root is safe for the second seat, so the proof works for either seat.

Every public pair overlaps. Otherwise the earlier-opened endpoint would CLOSE while its mate remained untouched, giving live degree one in the matching G_0 . Each pair is therefore charged exactly once. Hence

$$\sigma = \sum_i y_i p_i + \sum_h y_{a_h} y_{b_h} + \sum_{j:x_j=1} q_j = Q(x)$$

by the adapted-basis expansion.

The exact clock $2|U| + |L|$ drops by one on every OPEN and CLOSE. A nonterminal core never stalls: OPEN exists when $U \neq \emptyset$, and the exhausted-board CLOSE exists when $U = \emptyset \neq L$. Thus a board with N coins has exactly $2N$ core moves, after which it is the first player's turn. If $Q(x) = 1$, the first player follows the policy, forces $\sigma = 1$, and takes the tail. If $Q(x) = 0$, the second player follows the same policy, forces $\sigma = 0$, and leaves the first player without a move. This proves Theorem 1. The statement is strategic: it does not assert that arbitrary off-policy plays have charge $Q(x)$.

For F1, take the refinement-blind position universe to contain all coherent core states for both charge bits plus the sink; the fixed terminal set is the sink together with drained charge-zero states. Only a source OPEN whose mate is queued queries $Q(e_j)$, once and at weight one, proving F2/N2 at $(1, 1)$. Ordinary normal play is F3, and the proof is uniform over the full refinement torsor, giving N1. The empty input is already an optionless P -position because $Q(0) = 0$.

Remark 5 (contract-only shortcut). The written per-transition contract alone also admits a forced scan path: query the active singletons one per move, accumulate $P_B(x) + \sum_{j:x_j=1} Q(e_j) = Q(x)$, and parity-correct the path at the end. This meets the observation lower bound but is a distributed evaluator. The pass-free FIFO theorem is the stronger intended result: one public policy forces play-dependent overlap interactions for the entire refinement torsor. Thus access metering controls information but does not by itself axiomatize non-evaluative structure.

The pass-free transition, exact clock, safe-front strategy semantics, and impartial tail compiler are kernel-checked without axioms in `formal/Ogdoad/ImpartialRealizer.lean`. The weighted source loading and adapted-basis composition remain the explicit paper-level synthesis. The complete theorem has been folded into the flagship paper [1].

References

- [1] a9lim, *Quadratic refinements in normal play: realization and the observation boundary*, companion flagship paper, `writeups/goldarf.tex` in the ogdoad repository, 2026.
- [2] E. R. Berlekamp, J. H. Conway, R. K. Guy, *Winning Ways for Your Mathematical Plays*, 2nd ed., A K Peters, 2001–2004.
- [3] T. S. Ferguson, *Game Theory*, Part I, Ch. 5: Coin Turning Games, UCLA online lecture notes.
- [4] T. E. Plambeck, A. N. Siegel, *Misere quotients for impartial games*, J. Combin. Theory Ser. A **115** (2008), 593–622.